



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2024–25
PRE BOARD EXAMINATION

CLASS: X
SUBJECT: MATHEMATICS [SET A]

FULL MARKS: 80
TIME: 3 HOURS

Answers to this Paper must be written on a paper provided separately.

You will not be allowed to write during the first 15 minutes.

This time is to be spent in reading the question paper.

The time given at the head of this Paper is the time allowed for writing the answers.

Attempt all questions from Section A and any four questions from Section B.

The intended marks for questions or parts of questions are given in brackets [].

This paper consists of 9 printed pages.

SECTION A

(Attempt **all** questions from this section)

Question 1

Choose the correct answers to the questions from the given options. (Do not copy the question, Write the correct answer only.) [15]

(i) The SGST paid by a customer to the shopkeeper for an article, which is priced at ₹ 1000 is ₹ 60. The rate of GST charged is:

- (a) 5% (b) 12% (c) 18% (d) 28%

(ii) In solving a quadratic equation, one of the values of the variable x is 433.356. The solution rounded to two significant figures is:

- (a) 433.36 (b) 433.35 (c) 433.3 (d) 430

(iii) For the matrix product $A = \begin{bmatrix} 7 & -1 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} -2 & 5 \\ 3 & 8 \end{bmatrix}$, then which expression gives the value of A_{22} ?

- (a) $4(-2) + 2(3)$ (b) $7(5) + (-1)8$ (c) $4(5) + 2(8)$ (d) $(-1)3 + 2(8)$

(iv) If $x^{140} + 2x^{151} + k$ is divisible by $x + 1$, then the value of k is:

- (a) 1 (b) -3 (c) 2 (d) -2

(v) For the following distribution:

Class Interval	0 - 10	10 -20	20 -30	30 - 40	40 - 50
Frequency	20	30	24	40	17

The sum of the upper limits of the modal class and the median class is:

- (a) 50 (b) 70 (c) 80 (d) 100

(vi) Which of the following equation represent a line passing through origin?

- (a) $3x - 2y + 5 = 0$ (b) $2x - 3y = 0$ (c) $x = 5$ (d) $y = -6$

(vii) The straight line distance between Kolkata and a certain resort in Jharkhand was shown on the map as 12.7cm. If the scale is 2cm : 50 km the actual distance in km is:

- (a) 317.5km (b) 327 km (c) 327.5 km (d) 337km

(viii) Two right circular cones x and y are made, x having three times the radius of y and y having half the volume of x. The ratio of heights of x and y is:

- (a) 1: 4 (b) 1 : 7 (c) 2 : 3 (d) 2: 9

(ix) The given table shows the distance covered and time taken by a train moving at a uniform speed along a straight track.

Distance(in m)	80	120	y
Time(in sec)	4	x	5

The values of x and y are:

- (a) $x = 4, y = 150$ (b) $x = 6, y = 100$ (c) $x = 4, y = 100$ (d) $x = 3, y = 150$

(x) If a die is rolled, what is the probability of getting an odd number or a number greater than 4?

- (a) $\frac{1}{2}$ (b) $\frac{2}{3}$ (c) $\frac{3}{4}$ (d) 1

(xi) A lady intended to open a recurring deposit account of ₹1000 per month for 1 year in a bank, paying a 5% per annum rate of simple interest. The bank reduced the rate to 4% per annum. How much must she deposit monthly for 1 year so that her interest remains the same?

- (a) ₹1160 (b) ₹1200 (c) ₹1250 (d) ₹2000

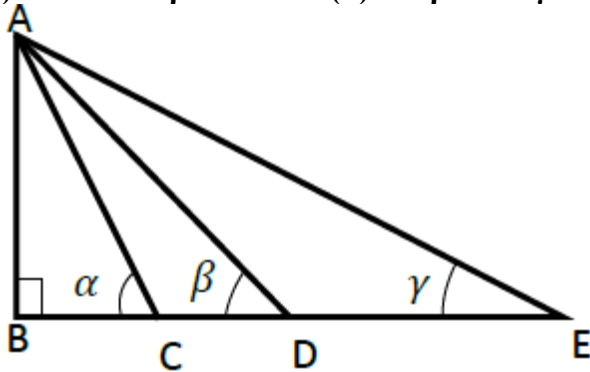
(xii) Mr X invests in ₹ 100, 12% shares of a Company A available at ₹60 each. Mr Y invests in ₹50, 16% shares of a Company B available at ₹ 40 each. Use this information to state which of the following statements is true.

- (a) The rate of return for Mr X is 12% (b) The rate of return for Mr Y is 10%
(c) Both Mr X and Mr Y have the same rate of return of 10%
(d) Both Mr X and Mr Y have the same rate of return of 20%

(xiii) An observer at point E, which is at a certain distance from the lamp post AB, finds the angle of elevation of top of lamp post from positions C, D and E as α , β and γ . It is given that B, C, D and E are along a straight line.

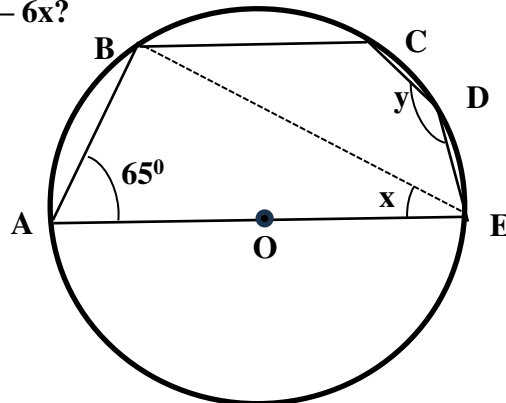
Which of the following condition is satisfied?

- (a) $\tan \alpha > \tan \beta$ (b) $\tan \beta < \tan \gamma$ (c) $\tan \gamma > \tan \alpha$ (d) $\tan \alpha < \tan \beta$



(xiv) In the given figure, O is the centre of the circle. Given that $BC \parallel AE$.

What is the value of $y - 6x$?



Statement 1 : The value $y - 6x$ is 5° .

Statement 2 : The value of angle y is 65° and x is 10° .

Which of the following is valid?

- (a) Both statements are true.
 (b) Both statements are false.
 (c) Statement 1 is true and Statement 2 is false
 (d) Statement 1 is false and Statement 2 is true

(xv) Assertion (A) : The point $(-3, 7)$ is invariant under the reflection $x = -3$

Reason (R) : If a point has its x-coordinate 0, it is invariant under reflection in both axes.

- (a) Both A and R are true, and R is the correct reason for A.
 (b) Both A and R are true, and R is incorrect reason for A.
 (c) A is true, R is false.
 (d) A is false, R is true.

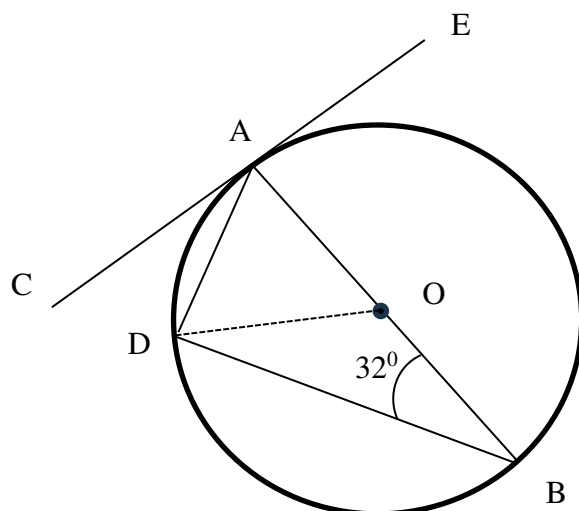
Question 2

(i) An engineering student was asked to make a model in her workshop, which was shaped like a cylinder with two cones attached to its two ends, using thin aluminium sheet. The diameter of the model is 3 cm and its length is 12 cm. If each cone has a height of 2 cm, find the volume of air contained in the model that she made. [4]

(ii) Find the equation of the line through the origin perpendicular to the line joining the points A (-1, -4) and B (7, 2). Also find the equation of the line through (3, 4) and parallel to AB. [4]

(iii) In the given figure, O is the centre of the circle. CE is a tangent to the circle at A. If $\angle ABD = 32^\circ$, then find:

- (a) $\angle BDA$
- (b) $\angle BAD$
- (c) $\angle CAD$
- (d) $\angle ODB$



[4]

Question 3

(i) Find the middle term of the sequence formed by all 3- digit numbers which leave a remainder 3, when divided by 4. Also find the sum of all numbers on both sides of the middle term separately. [4]

(ii) A man deposited ₹600 per month in a recurring deposit account. The bank pays a simple interest of 12% p.a. Calculate the:

- (a) number of monthly instalments he deposits to get a maturity amount of ₹11826?
- (b) total interest paid by the bank.
- (c) total amount deposited by him. [4]

(iii) Use graph sheet to answer this question. (Take 1 cm = 1 unit along both x and y axes). Plot A(0, 3), B(4, 0), C(6, 2) and D(5, 0). Reflect the points as given below and write their coordinates:

- (a) Reflect A on x- axis and name it as A'.
- (b) Reflect B on y- axis and name it as B'.
- (c) Reflect C on x- axis and name it as C'.
- (d) D remains invariant when reflected on the line whose equation is -----.
- (e) Join points A, B, C, D, C', B, A', B', and A to form a closed figure.

Give a geometrical name of the closed figure BCDC'.

[5]

SECTION B

(Attempt **any four** questions from this section)

Question 4

(i) The marks scored by 100 students are given below:

Marks scored	0 -10	10 -20	20- 30	30 -40	40- 50	50- 60	60- 70	70 -80	80 -90	90- 100
No. of students	4	5	7	9	12	13	11	15	10	14

A student in the class is selected at random. Find the probability that the student has scored:

(a) atleast 20

(b) below 60 but 30 or more

(c) atmost 100.

[3]

(ii) If p, q, r are in continued proportion, then prove that,

$$\frac{3p^2 + 5pq + 7q^2}{3q^2 + 5qr + 7r^2} = \frac{p}{r}$$

[3]

(iii)(a) A rectangular water tank of base 7m × 6m contains water upto a depth of 5m. How many cubic metres of water is there in the tank?

(b) If the water in the tank in (a) is transferred to a cylindrical tank of radius 3.5m, how high will the water level be in the tank? (Give your answer correct to one place of decimal).

(c) A conical solid of base radius 2.8m is put into the tank referred in (b) above then the water level is raised by 0.16m. Calculate the height of the conical solid.

[4]

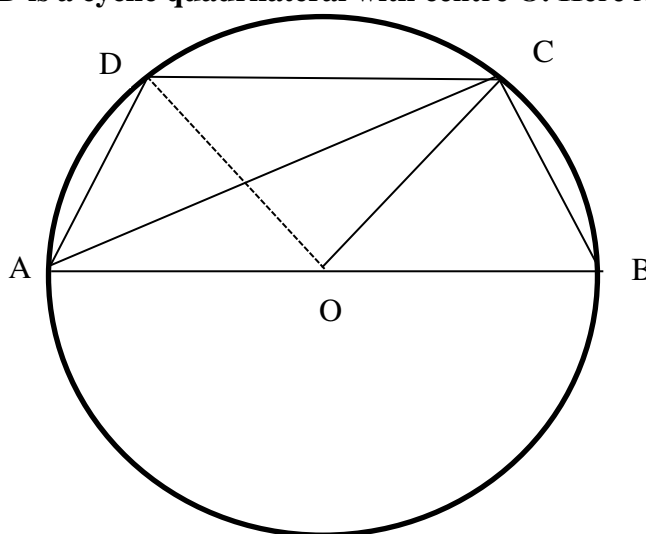
Question 5

(i) In the given circle ABCD is a cyclic quadrilateral with centre O. Here AB ∥ DC and ∠CAB = 25°. Find:

(a) ∠COB

(b) ∠ADC

(c) ∠DAC



[3]

(ii) The following bill shows the GST rate and the marked price of articles:

ABP Groceries				
S. No:	Item	Marked Price	Discount	Rate of GST
(a)	Kelloggs Muesli(Fruit and Nut)	₹ 450	15%	18%
(b)	Daawat Biryani Rice(1 kg)	₹220	Nil	5%
(c)	Govardhan Ghee (500 ml)	₹ 300	Nil	12%

Find the total amount to be paid (including GST) for the above bill.

[3]

(iii) In the given figure (not drawn to scale), BC is parallel to EF, CD is parallel to FG, $AE : EB = 2 : 3$, $\angle BAD = 70^\circ$, $\angle ACB = 105^\circ$, $\angle ADC = 40^\circ$ and AC is bisector of $\angle BAD$.

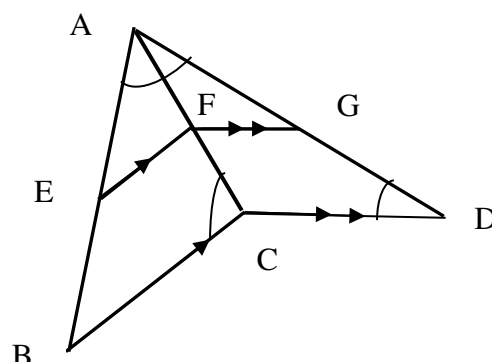
(a) Prove $\triangle AEF \sim \triangle AGF$

(b) Find:

(i) $AG : AD$

(ii) Area of $\triangle ACB : \triangle ACD$

(iii) Area of quadrilateral ABCD : area of $\triangle ACB$



[4]

Question 6

(i) Solve the quadratic equation $2x^2 - 5 = 10x$ for x and express your answer correct to two places of decimal.

[3]

(ii) The following table gives the price of diesel per litre for a period of 50 days.

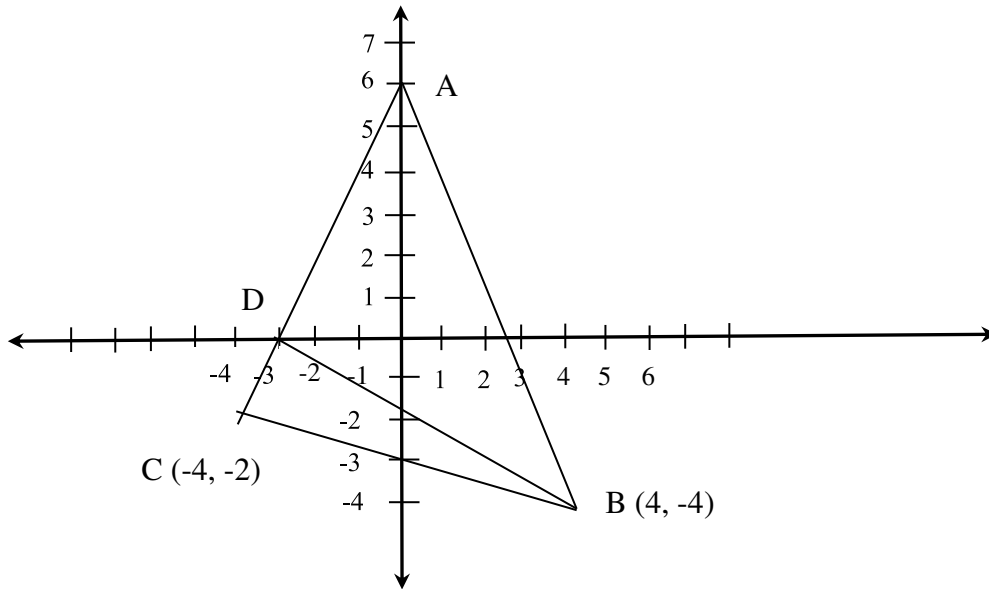
Price(₹)	85 -90	90 - 95	95 -100	100 - 105	105 - 110
No: of days	12	10	8	15	5

Find the mean price of diesel per litre to the nearest rupee using step-deviation method.

[3]

(iii)

In the given diagram, ABC is a triangle, where B (4, -4) and C (-4, -2). D is a point on AC.



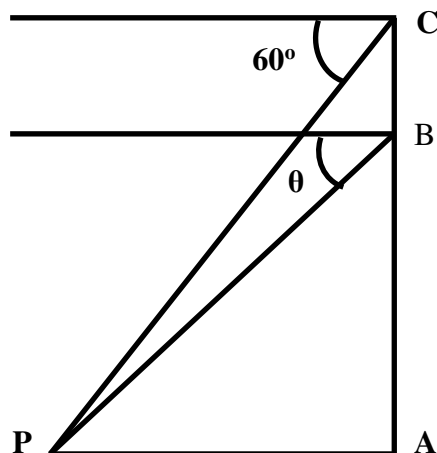
- (a) Write down the coordinates of A and D.
- (b) Find the coordinates of the centroid of $\triangle ABC$
- (c) If D divides AC in the ratio $k : 1$, find the value of k
- (d) Find the equation of the line BD.

[4]

Question 7

(i) A hot air balloon is rising vertically from a point A on the ground which is at a distance of 50 m from a car parked at a point P on the ground. A man who is riding the balloon observes that it took him 10 seconds to reach a point B which he estimated to be equal to the horizontal distance of his starting point from the car parked at P.

- (a) Find the angle of depression " θ " from the balloon to the car parked at P.
- (b) Find the speed of the hot air balloon.
- (c) After a certain time, on reaching point C, he observes that the angle of depression is equal to 60° . Calculate the vertical distance travelled by the balloon during this time.
- (d) Find the total time taken by the balloon to reach the point C from the ground.
- (e) Find the straight line distance of the line joining point C from the car parked at P.



[5]

(ii) The following data represents the daily wages in rupees of a certain number of employees of a company:

Daily wages(in ₹)	30 - 40	40 - 50	50 - 60	60 - 70	70 - 80	80 - 90	90 - 100	100 - 110
No: of employees	8	14	12	17	20	26	13	10

On a graph sheet draw an ogive using the above data. Take 2cm = ₹10 along one axis and 2cm = 20 employees along the other axis. Use your graph to find the:

- (a) median
 (b) upper quartile
 (c) number of employees who earn more than ₹ 84 per day. [5]

Question 8

(i) Solve the following inequation, write down the solution set and represent it on the real number line.

$$\frac{1}{2} (2x - 1) \leq 2x + \frac{1}{2} < 5 \frac{1}{2} + x, x \in W \quad [3]$$

(ii) A man sells 2000 Himalaya Tea (₹ 100) ordinary shares, which pay a dividend of 2.5% at ₹33 per share. He invests the proceeds in Cotton Textiles (₹ 25) ordinary shares, at ₹ 44 per share which pay a dividend of 15%. Find:

- (a) the number of Cotton Textile shares purchased.
 (b) the change in his dividend income (assume that there is no brokerage on either transaction). [3]

(iii) If $A = \begin{bmatrix} -1 & 3 \\ 2 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 1 & -2 \\ 0 & 3 \end{bmatrix}$, $C = \begin{bmatrix} 1 & -4 \end{bmatrix}$ and $D = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$.

- (a) Is the product AC possible? Justify your answer.
 (b) Find the matrix X, such that $X = AB + B^2 - DC$ [4]

Question 9

(i) The following distribution gives the daily wages of 60 workers of a factory.

Daily income in ₹	200-300	300 - 400	400 - 500	500 - 600	600 - 700	700 - 800
Number of Workers (f)	6	10	14	16	10	4

Use graph paper to answer this question.

Take 2cm = ₹100 along one axis and 2cm = 2 workers along the other axis.

Draw a histogram and hence find the mode of the given distribution. [3]

(ii) Find the Geometric Progression for which the sum of first two terms is -4 and the fifth term is 4 times the third term. [3]

- (iii) A student factorized the following polynomial : $x^3 + 10x^2 - 37x + 26$. She found the result as $(x + 1)(x - 2)(x + 3)$, using remainder and factor theorem, verify whether her result is correct. If incorrect, give the correct result. [4]

Question 10

- (i) The total expenses of a trip for a certain number of people is ₹18000. If three more people join them, then the share of each reduces by ₹3000. Take “x” to be the original number of people, form a quadratic equation in “x” and solve it to find the value of x. [3]
- (ii) Prove the following identity:

$$\frac{(1 + \cot A + \tan A)(\sin A - \cos A)}{\sec^3 A - \operatorname{cosec}^3 A} = \sin^2 A \cos^2 A \quad [3]$$

- (iii) Use ruler and compasses only for this question. Construct a circle of radius 4.5cm. Draw a chord AB = 6cm.
- (a) Find the locus of the points equidistant from A and B. Mark the point where it meets the circle as D.
- (b) Join AD and find the locus of points which are equidistant from AD and AB. Mark the point where it meets the circle as C.
- (iii) Join BC and CD. Measure and write down the length of side CD of the quadrilateral ABCD. [4]